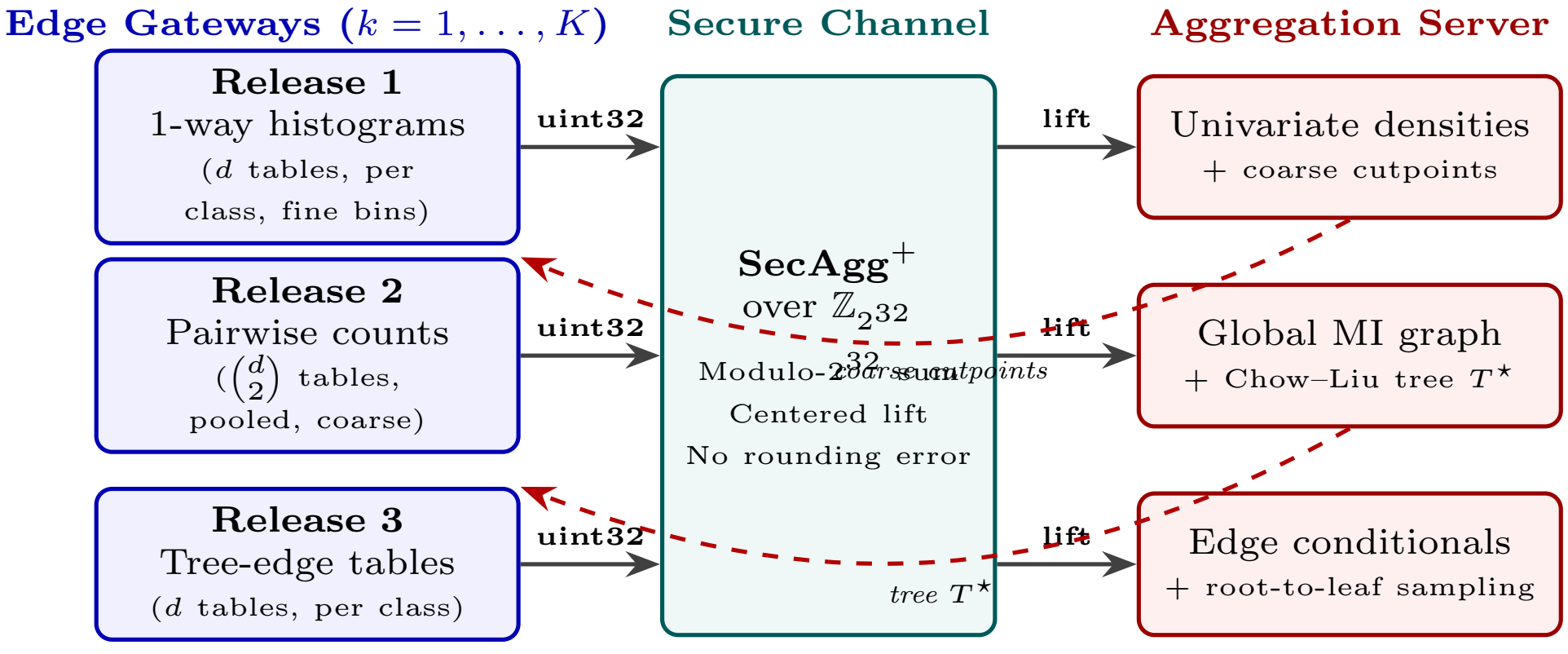

Edge Gateways ($k = 1, \ldots, K$)
Secure Channel
Aggregation Server
Release 1
1-way histograms
(d tables, per class, fine bins)
Release 2
Pairwise counts
($\binom{d}{2}$ tables, pooled, coarse)
Release 3
Tree-edge tables
(d tables, per class)
uint32
uint32
uint32
SecAgg$^+$
over $\mathbb{Z}_{2^{32}}$
Modulo-2^{32} sum
Centered lift
No rounding error
coarse cutpoints
tree $T^\star$
lift
lift
lift
Univariate densities
+ coarse cutpoints
Global MI graph
+ Chow–Liu tree $T^\star$
Edge conditionals
+ root-to-leaf sampling